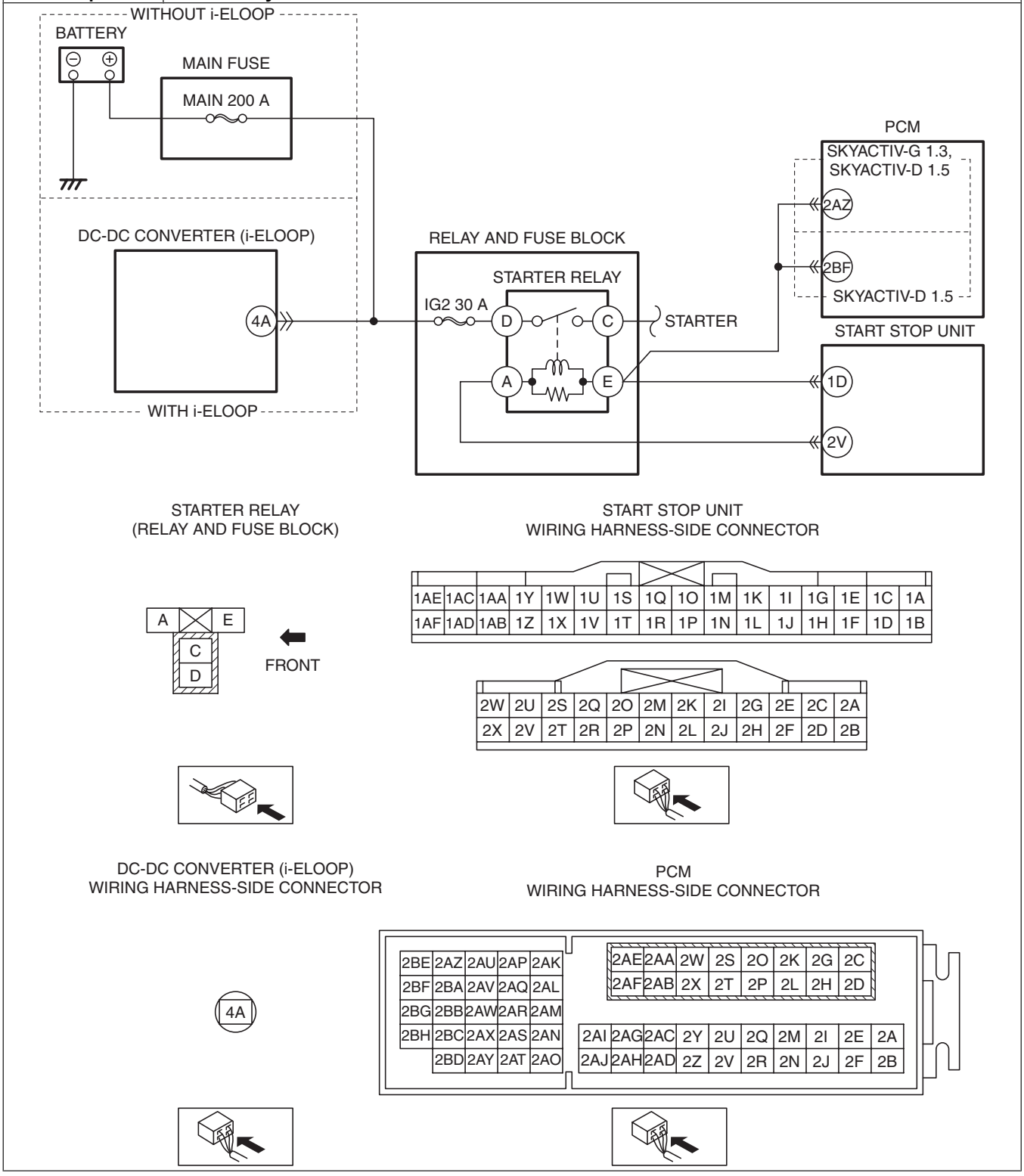
DTC P0615:12 [START STOP UNIT]

id0902p6022100

Description	Starter relay circuit malfunction
Detection condition	• Start stop unit detects that the starter relay circuit voltage of specification or more for **0.1 s or more** with the ignition switched off.
Fail-safe	• Inhibits switching the ignition ON (engine off or on). • Inhibits the engine start.
Possible cause	• DTCs are stored in the PCM. • Starter relay malfunction • Start stop unit connector or terminal malfunction • PCM connector malfunction • Short to power supply in wiring harness between the following terminals: — Start stop unit terminal 1D—Starter relay terminal E — Starter relay terminal E—PCM terminal 2AZ (SKYACTIV-G 1.3, SKYACTIV-G 1.5) — Starter relay terminal E—PCM terminal 2BF (SKYACTIV-D 1.5) — Start stop unit terminal 2V—Starter relay terminal A • PCM malfunction • Start stop unit malfunction

Description	Starter relay circuit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT STARTER RELAY** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the starter relay. (See RELAY LOCATION.) • Inspect the starter relay. (See RELAY INSPECTION.) • Is the starter relay normal?	Yes	Go to the next step.
		No	Replace the starter relay, then go to Step 7. (See RELAY LOCATION.)
3	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
5	**INSPECT STARTER RELAY CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the starter relay is removed. • Verify that the start stop unit and PCM connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the following terminals (wiring harness-side): — Start stop unit terminal 1D — Start stop unit terminal 2V • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Start stop unit terminal 1D—Starter relay terminal E • Starter relay terminal E—PCM terminal 2AZ (SKYACTIV-G 1.3, SKYACTIV-G 1.5) • Starter relay terminal E—PCM terminal 2BF (SKYACTIV-D 1.5) • Start stop unit terminal 2V—Starter relay terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 7.
6	**VERIFY IF MALFUNCTIONING LOCATION IS PCM DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition off and wait for **1 s or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to Step 8.

Step	Inspection		Action
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition off and wait for **1 s or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.